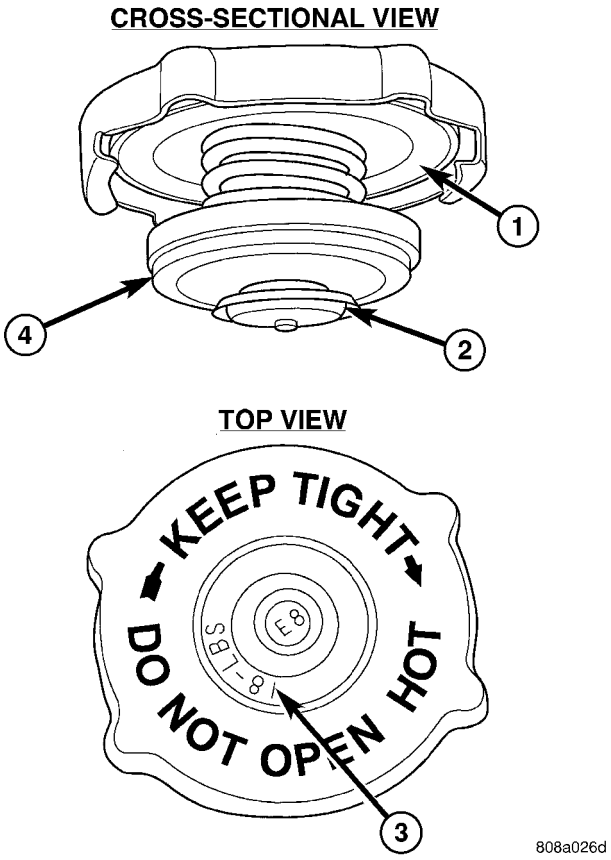

Fig. 34 Radiator Pressure Cap - Typical

1 - FILLER NECK SEAL
2 - VACUUM VENT VALVE
3 - PRESSURE RATING
4 - PRESSURE VALVE

OPERATION

A vent valve in the center of the cap will remain shut as long as the cooling system is pressurized. As the coolant cools, it contracts and creates a vacuum in cooling system. This causes the vacuum valve to open and coolant in reserve/overflow tank to be drawn through connecting hose into radiator. If the vacuum valve is stuck shut, or overflow hose is kinked, radiator hoses will collapse on cool-down.

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - RADIATOR CAP - TO - FILLER NECK SEAL

The pressure cap upper gasket (seal) pressure relief can be tested by removing overflow hose from radiator filler neck nipple. Attach hose of pressure tester tool 7700 (or equivalent) to nipple. It will be necessary to disconnect hose from its adapter for filler neck. Pump air into radiator. The pressure cap upper gasket should relieve at 69-124 kPa (10-18 psi) and hold pressure at a minimum of 55 kPa (8 psi).

WARNING: The warning words - DO NOT OPEN HOT - on radiator pressure cap, are a safety precaution. when hot, pressure builds up in cooling system. To prevent scalding or injury, radiator cap should not be removed while system is hot and/or under pressure.

Do not remove radiator cap at any time **except** for the following purposes:

(1) Check and adjust antifreeze freeze point.
(2) Refill system with new antifreeze.
(3) Conducting service procedures.
(4) Checking for vacuum leaks.

WARNING: If vehicle has been run recently, wait at least 15 minutes before removing radiator cap. With a rag, squeeze radiator upper hose to check if system is under pressure. Place a rag over cap and without pushing cap down, rotate it counter-clockwise to first stop. Allow fluid to escape through the coolant reserve/overflow hose into reserve/overflow tank. Squeeze radiator upper hose to determine when pressure has been released. When coolant and steam stop being pushed into tank and system pressure drops, remove radiator cap completely.

DIAGNOSIS AND TESTING - RADIATOR CAP

Remove cap from radiator. Be sure that sealing surfaces are clean. Moisten rubber gasket with water and install cap on pressure tester 7700 or an equivalent (Fig. 35).

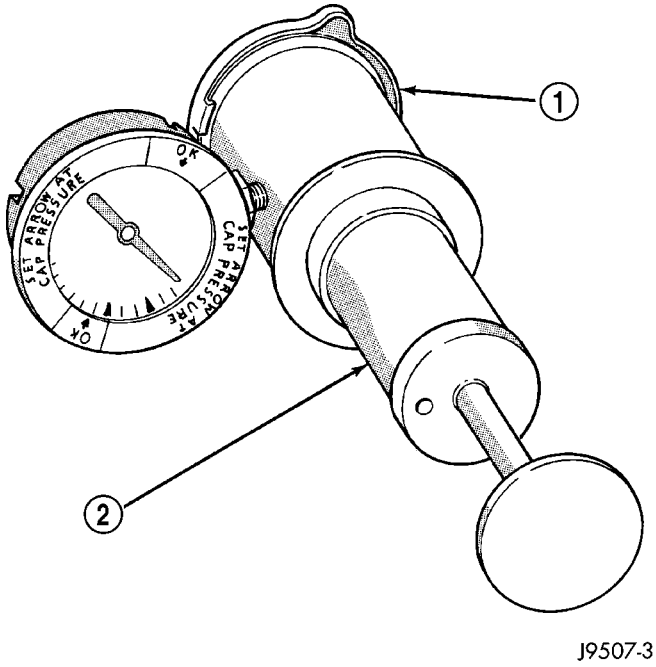

Fig. 35 Pressure Testing Radiator Pressure Cap—Typical

1 - PRESSURE CAP
2 - TYPICAL COOLING SYSTEM PRESSURE TESTER